The Aqua Cycle
Go Green, Go Fresh

BUSINESS PROPOSAL

TEAM 4

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PLASTIC
IS CREATING A 326 YEAR PROBLEM

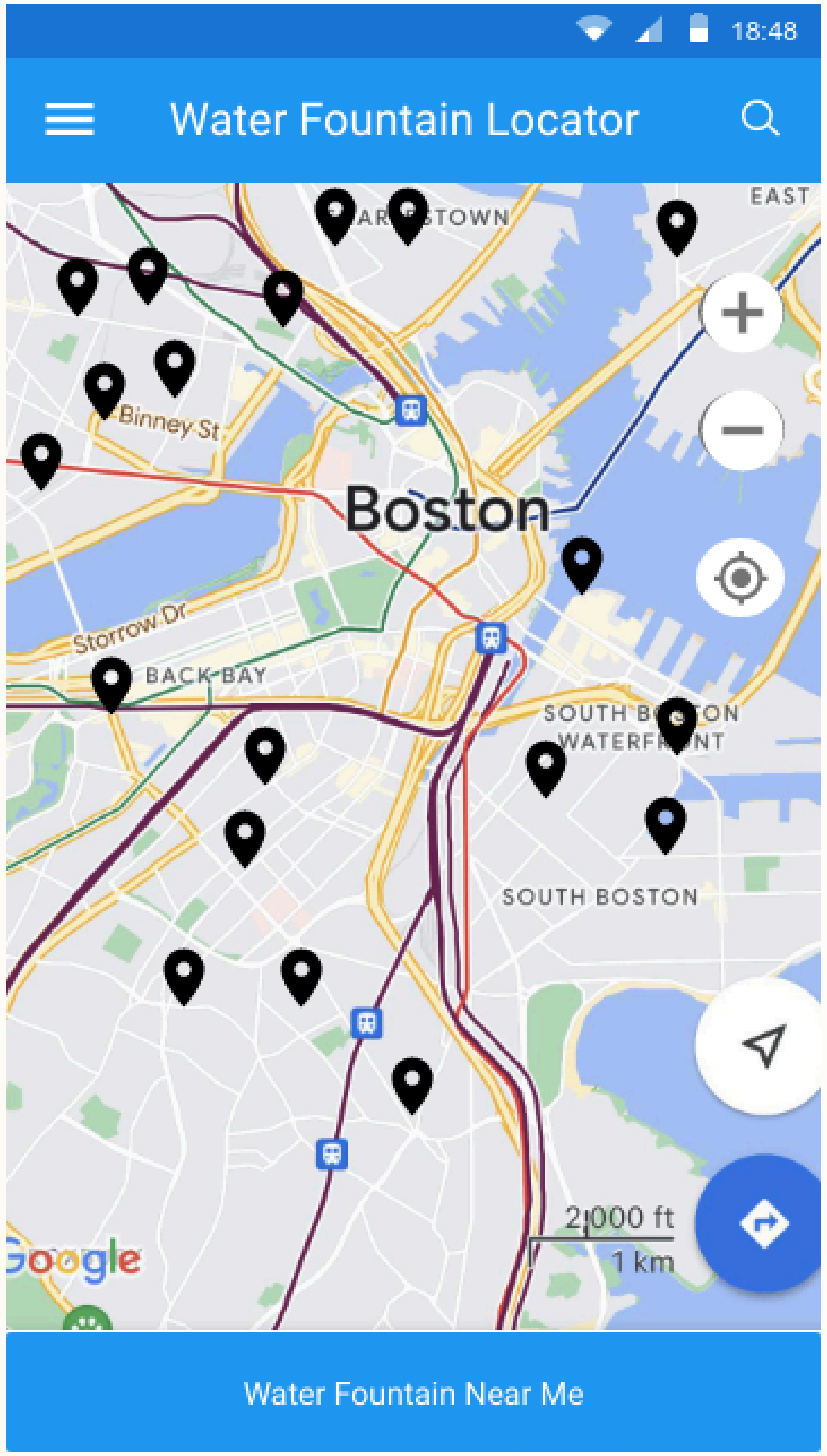
MISSION

Our mission is to reduce plastic waste and promote sustainable living by providing accessible, IoT-enabled water fountains integrated with an intuitive app. Through innovation, community engagement, and environmental responsibility, we aim to inspire individuals to adopt eco-friendly habits and contribute to a healthier planet.

VISION

We envision a world where sustainable hydration is the norm, with widespread access to smart water fountains that eliminate the need for single-use plastic bottles. By combining technology, education, and environmental awareness, the Aqua Cycle Project seeks to create a lasting positive impact on global ecosystems and future generations.

The Aqua Cycle
Go Green, Go Fresh
This app connects users to public water fountains, reducing single-use plastics and promoting community sustainability.
Log In
Sign Up



18:48
Water Fountain Locator
EAST
Binney St
Boston
Storrow Dr
BACK BAY
SOUTH BOSTON
2,000 ft
1 km
Google
Water Fountain Near Me

Hult House, Cambridge
MA< USA
Align the QR Code within frame to scan
OK

UNIQUE **BENEFITS**

"Our initiative tackles three key areas: environment, economy, and engagement. We're cutting single-use plastic waste, saving municipalities money through predictive maintenance, and gamifying user experience to ensure participation.

STRATEGIC **IMPORTANCE**

Sustainability Goals:
Direct alignment with global SDGs, including Clean Water (SDG 6) and Responsible Consumption (SDG 12).
Public-Private Collaboration:
Opportunities for partnerships with local governments, businesses, and NGOs.
Scalable Model:
Adaptable across urban centers globally for maximum impact

IMPACT **POTENTIAL**

Environmental Impact:
Significant reduction in single-use plastic waste by promoting reusable bottles.
Economic Efficiency: Cost savings for municipalities via predictive maintenance and reduced waste management expenses.
User-Centric Design: Gamified rewards, real-time fountain locators, and personalized dashboards drive engagement.

- **Pilot Program** (**0-6 Months**)

Deploy IoT-enabled smart fountains in 10 high-traffic locations (e.g., Boston downtown, campuses).

Develop MVP for the mobile app with real-time locators, sustainability dashboards, and rewards.

Gather Data: Collect usage patterns, foot traffic, and water quality data for refining AI models.

Phase 1

Phase 2

- **Regional Expansion** (**6-18 Months**)

Scale deployment to 20+ locations within the city, focusing on underserved neighborhoods.

Enhance Features: Add predictive maintenance and voice navigation capabilities.

Engage Stakeholders: Collaborate with schools, businesses, and NGOs for wider adoption.

Phase 3

- **National and Global Scaling** (**18+ Months**)

Expand to other cities and regions with customization for local contexts.

Integrate app features with public transportation and carbon footprint trackers.

Monitor & Optimize: Use real-time analytics for ongoing improvements and impact measurement.

COST

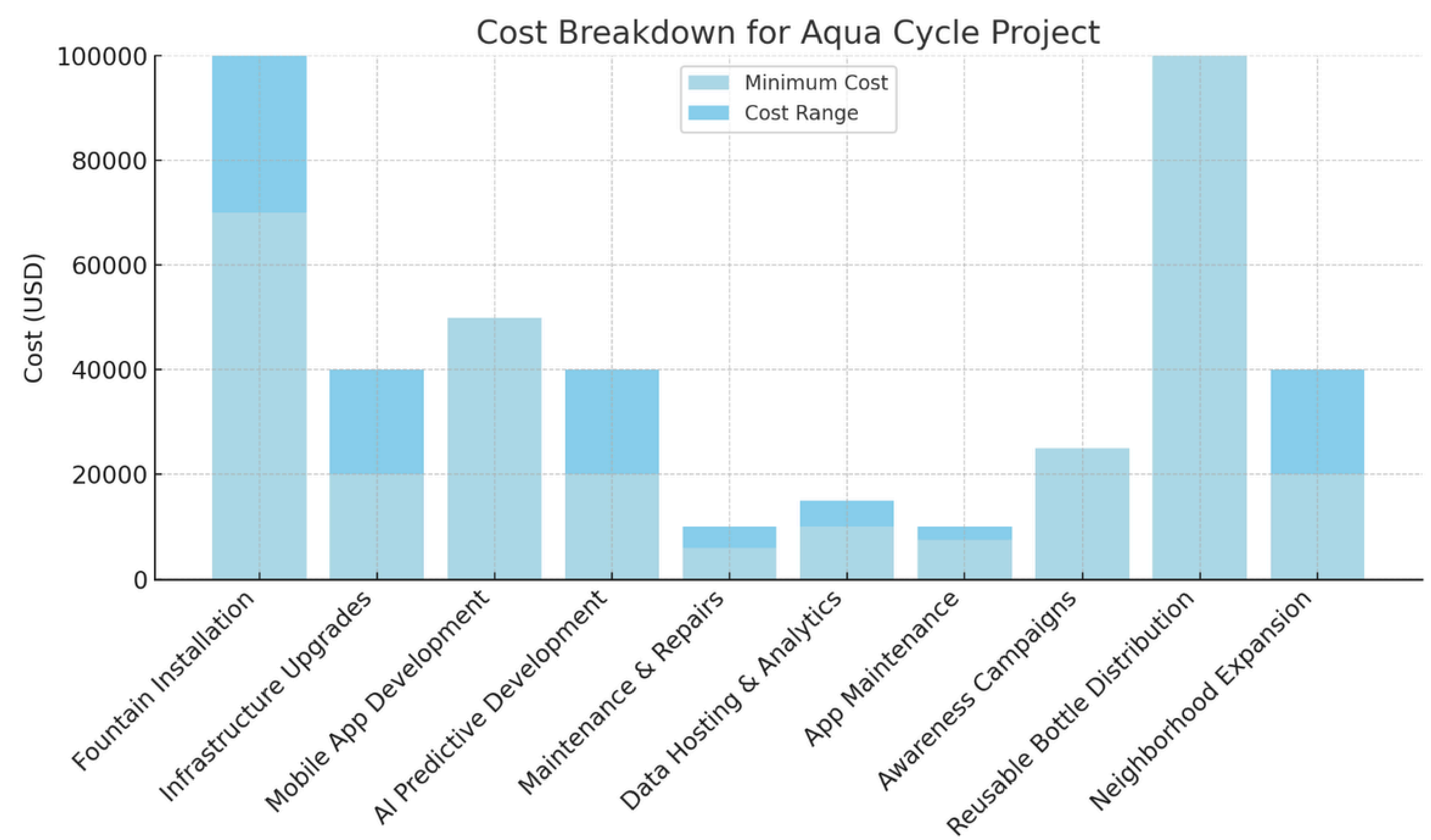
This chart illustrates the cost distribution for key components of the Aqua Cycle Project. The major costs are:

1. Fountain Installation (up to $100,000) and Reusable Bottle Distribution (up to $100,000) as primary cost drivers.

1. Infrastructure Upgrades and AI Predictive Development have mid-range costs between $40,000 and $120,000.

1. Operational Costs like Maintenance, Data Hosting, and App Maintenance remain smaller but ongoing expenses.

Total Cumulative Costs $98000 across all the phases



- **Pilot Program (0-6 Months)**

The initial costs are focused on Fountain Installation ($40,000 - $70,000) and Infrastructure Upgrades ($10,000 - $20,000). Additional costs include App Development ($120,000) and AI Predictive Model ($20,000). Maintenance costs for the fountains total $5,000 annually, while Community Engagement initiatives are budgeted at $10,000. The cost of Reusable Bottle Distribution adds $5,000.
Total Phase 1 Cost: ~$140,000 - $220,000

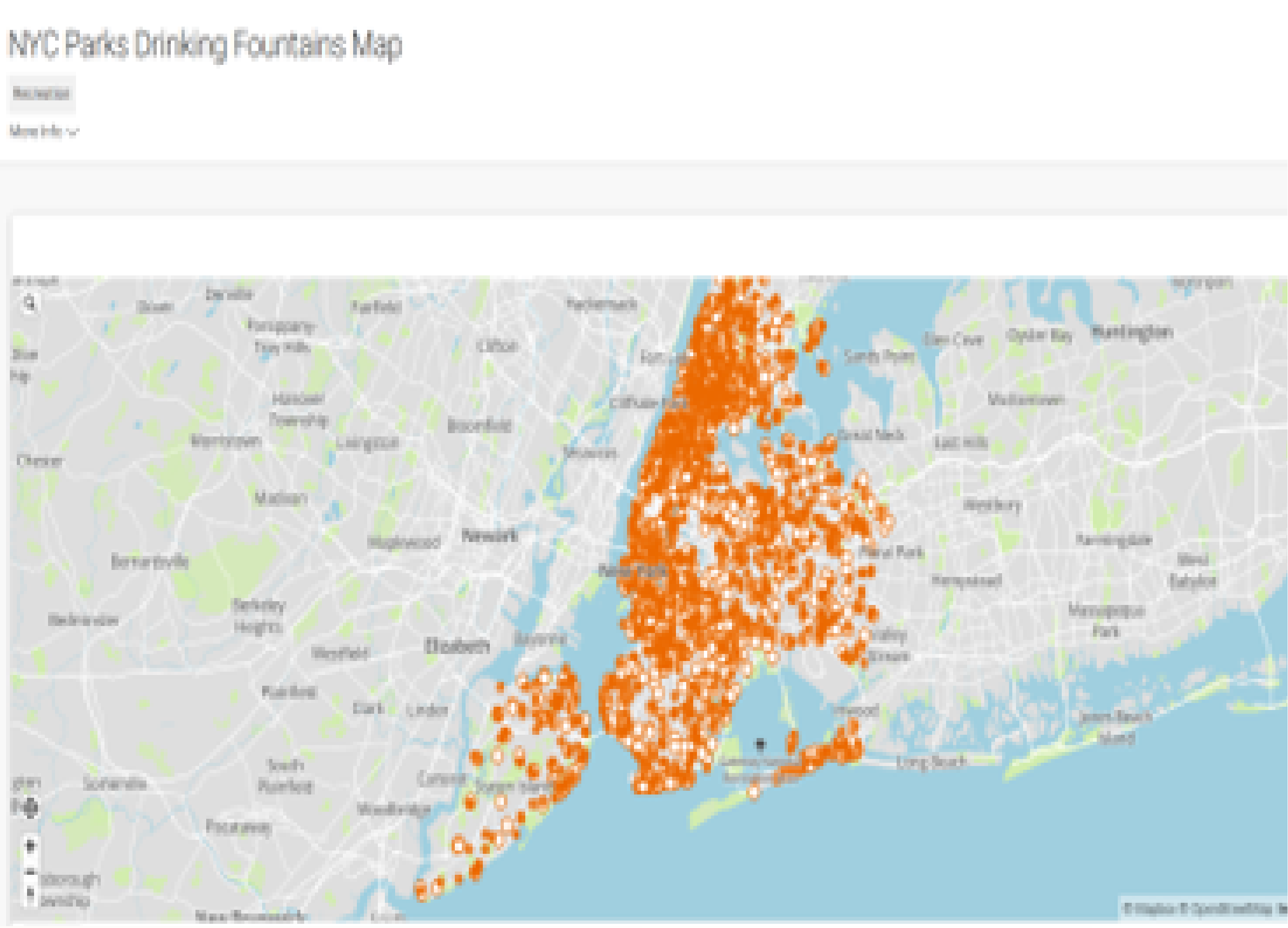
- **Regional Expansion (6-18 Months)**

The installation of 50–60 fountains will cost $200,000–$420,000, with infrastructure upgrades adding $50,000–$120,000. Additional costs include App Development ($300,000), AI Predictive Models ($20,000), and annual Maintenance ($25,000–$30,000). Community Engagement is budgeted at $20,000, and Reusable Bottle Distribution will cost $25,000–$30,000.
Total Phase 2 Cost: ~$690,000–$760,000.

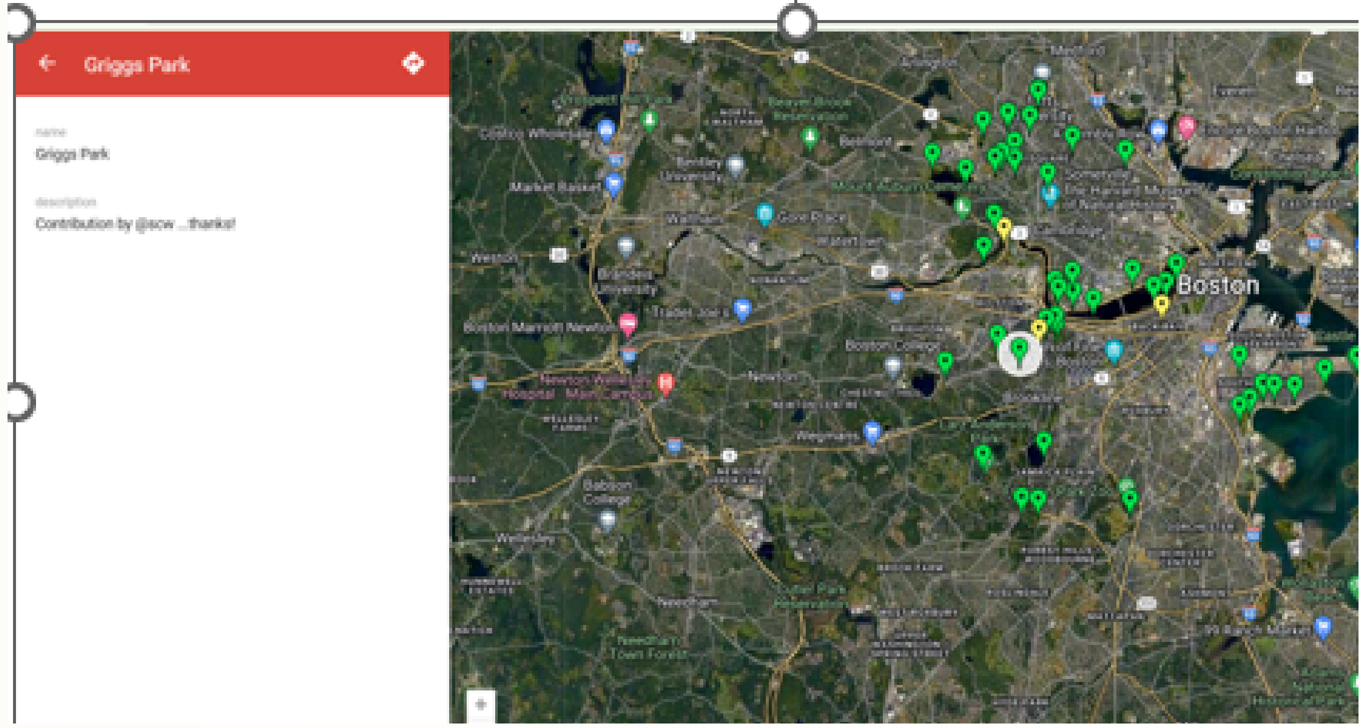
- **National and Global Scaling (18+ Months)**

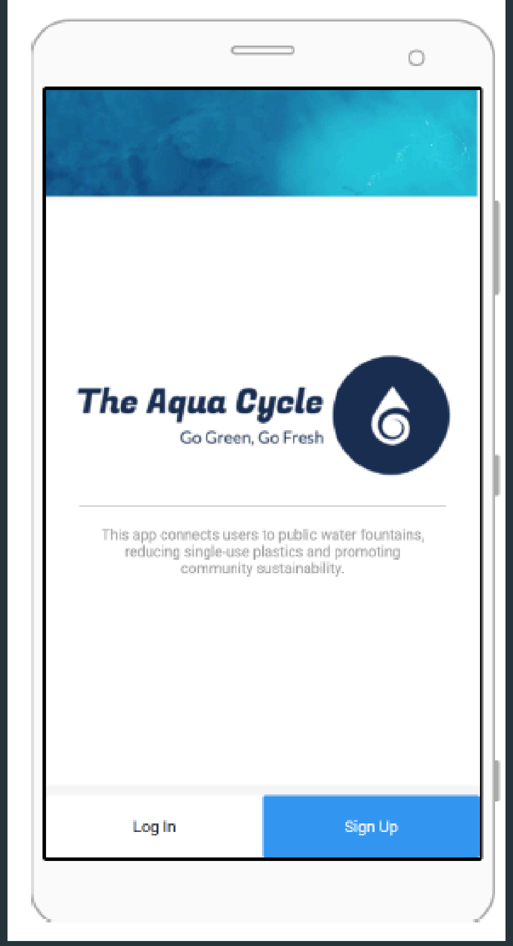
Cost Intensive but operational efficiency

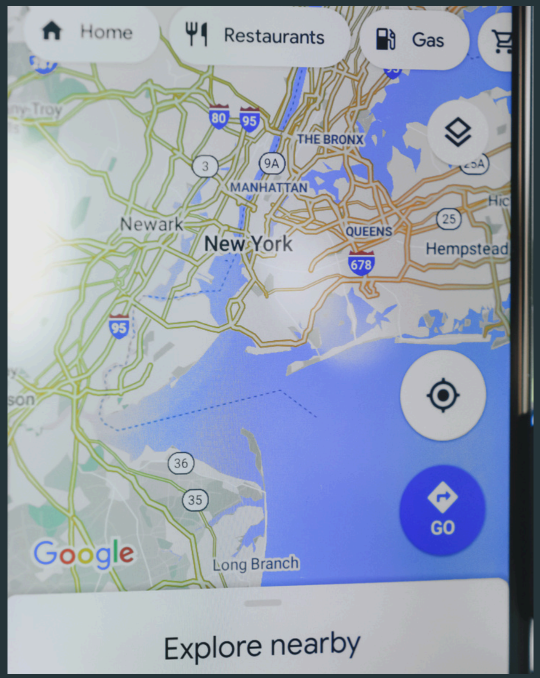
Concentration of normal water fountains in New York parks :



Concentration of drinkable fountains in Boston:







Investors

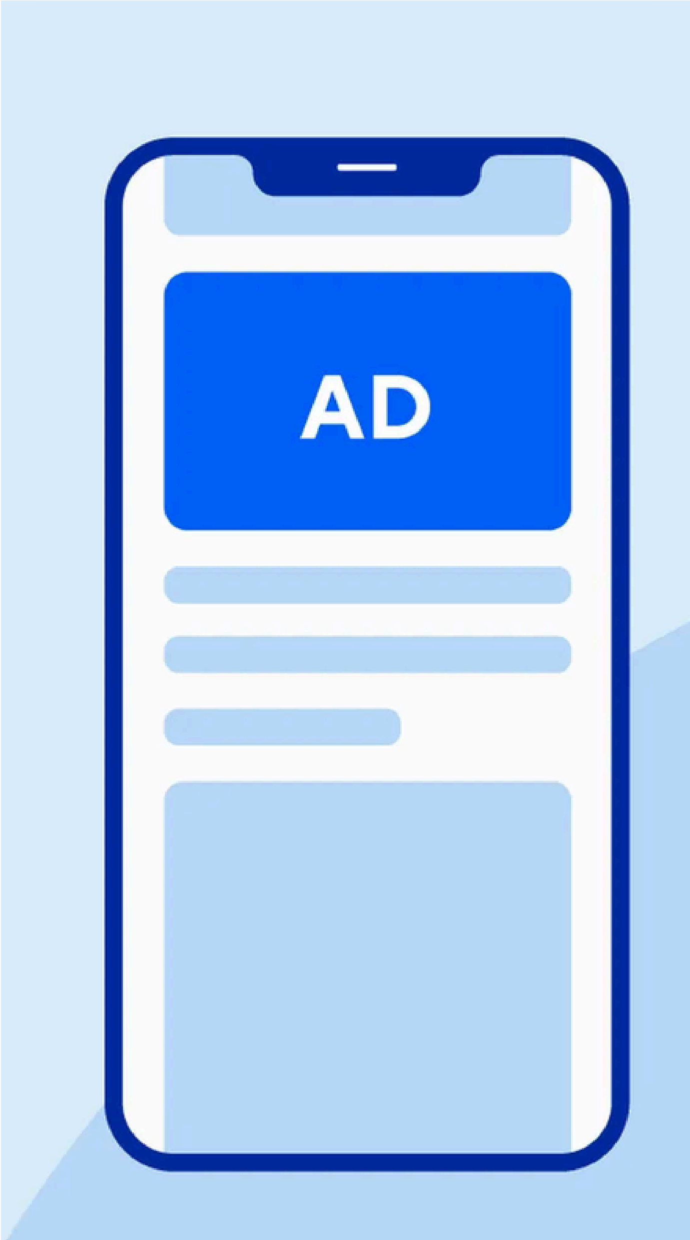
1) Company interested in CSR activities
2)Government
3)NGO

PROJECTED
REVENUE

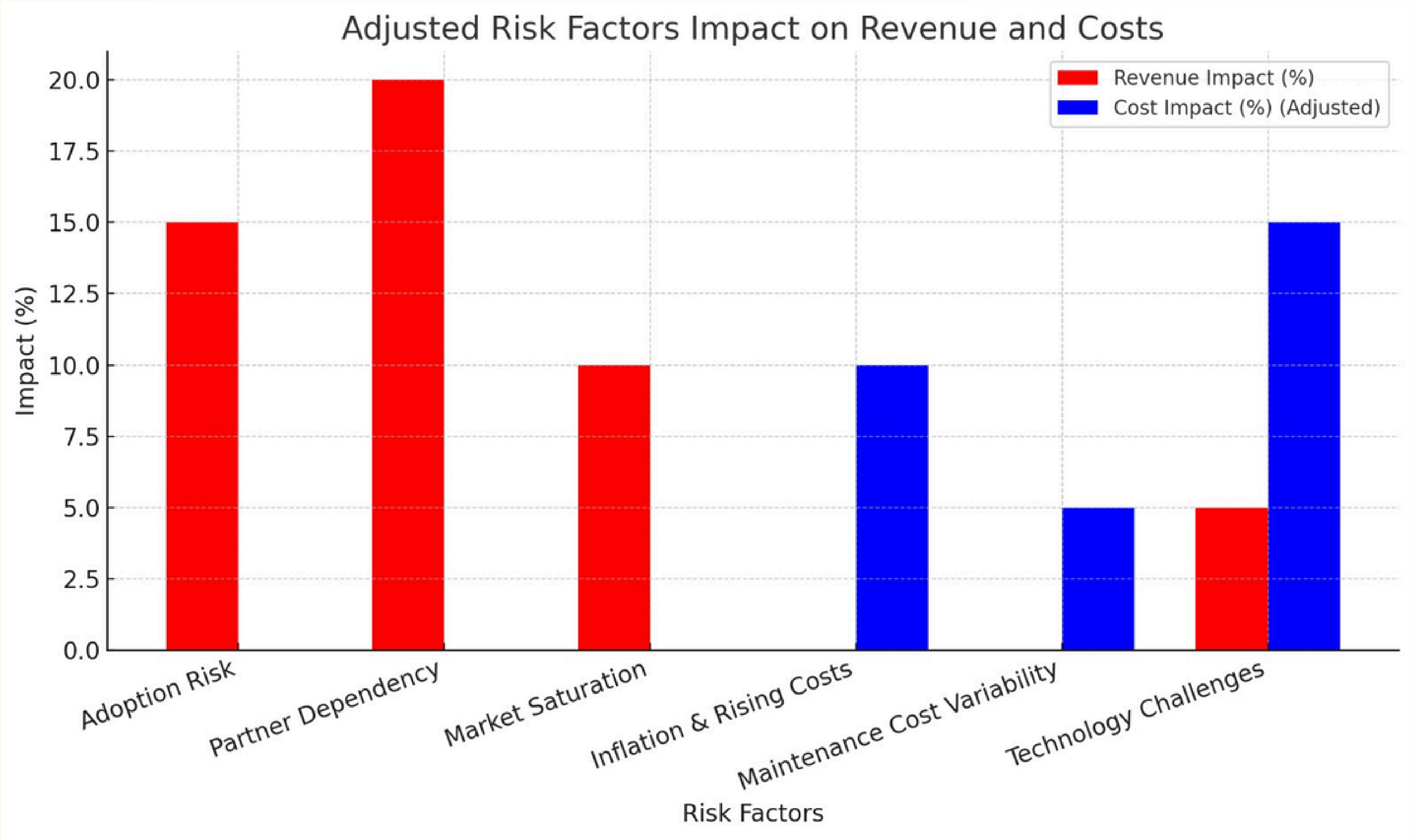
- Nominal User Fees (Charging 50 cents for one litre)

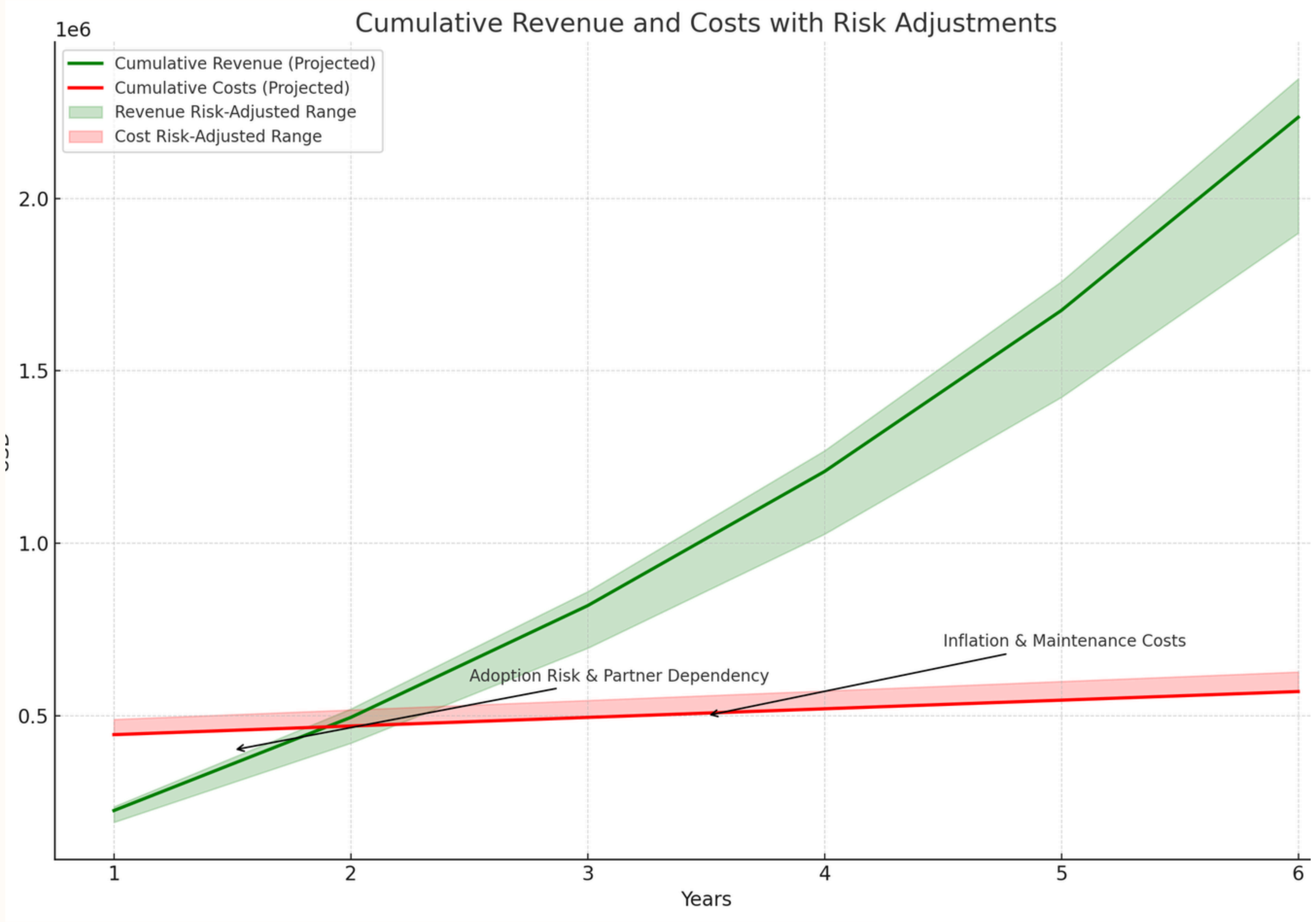
Estimate 100,000 Users in first 6 months
Estimating daily revenue of $50000.
In one year we estimate $18,250,000.00

- In Application Advertisements. (Boston: $10 up to $200 a day)



Disclaimer
The fees charged goes towards maintenance and to create value.


Adjusted Risk Factors Impact on Revenue and Costs
Revenue Impact (%)
Cost Impact (%) (Adjusted)
Impact (%)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Adoption Risk
Partner Dependency
Market Saturation
Inflation & Rising Costs
Maintenance Cost Variability
Technology Challenges
Risk Factors


Cumulative Revenue and Costs with Risk Adjustments
1e6
Cumulative Revenue (Projected)
Cumulative Costs (Projected)
Revenue Risk-Adjusted Range
Cost Risk-Adjusted Range
2.0
1.5
1.0
0.5
Inflation & Maintenance Costs
Adoption Risk & Partner Dependency
1
2
3
4
5
6
Years

CHALLENGES

TECHNICAL LIMITATIONS

·IOT SENSOR CALIBRATION: INCONSISTENT DETECTION OF WATER QUALITY AND USAGE PATTERNS.

·PREDICTIVE MAINTENANCE GAPS: LIMITED DATASETS CAUSING DELAYS AND INACCURATE ALERTS.

MARKET ADOPTION

·USER ENGAGEMENT: TESTERS FOUND THE APP INTERFACE CONFUSING; REQUESTED STREAMLINED FEATURES.

·BEHAVIOR RESISTANCE: USERS STRUGGLE TO SHIFT FROM SINGLE-USE PLASTICS DUE TO CONVENIENCE.

RESOURCE REQUIREMENTS

·HIGH COSTS: IOT-ENABLED SYSTEMS REQUIRE SIGNIFICANT UPFRONT INVESTMENT.

·DATA SUFFICIENCY: LARGER DATASETS ARE NEEDED FOR AI SYSTEMS TO OPTIMIZE AND DELIVER ACTIONABLE INSIGHTS.

SOLUTIONS

·ENHANCED SENSOR CALIBRATION AND EXTENDED PILOT TESTING FOR BETTER DATA ACCURACY AND PREDICTIVE MAINTENANCE.

·SIMPLIFIED APP INTERFACE WITH GAMIFIED REWARDS AND NOTIFICATIONS TO DRIVE USER ENGAGEMENT.

·PARTNERSHIPS WITH MUNICIPALITIES AND SPONSORS TO LOWER COSTS AND MAXIMIZE ROI.

PROJECTED IMPACT

·10% REDUCTION IN SINGLE-USE PLASTICS, PREVENTING 10,000+ POUNDS OF WASTE ANNUALLY IN BOSTON.

·20% INCREASE IN CLEAN DRINKING WATER ACCESS, ESPECIALLY IN UNDERSERVED NEIGHBORHOODS.

·SCALABLE AI-POWERED DEPLOYMENT MODEL FOR GLOBAL EXPANSION.

FUTURE ENHANCEMENTS AND SCALABILITY

Proposed Steps for Future Scaling

1. **Scalability:**
    - Deploy smart fountains in high-density urban areas to maximize visibility and user impact.
    - Leverage geospatial data and AI-driven insights to optimize fountain placement, particularly in underserved regions.
    - Expand globally by partnering with international organizations to address water scarcity challenges.
2. **Advanced AI Integration:**
    - Introduce adaptive algorithms for dynamic adjustments based on real-time environmental and usage data.
    - Enhance predictive maintenance models with extended datasets and continuous feedback loops.
    - Utilize machine learning to forecast demand and optimize water quality management.
3. **Enhanced User Engagement:**
    - Develop advanced gamification features like user competitions and shareable achievements.
    - Expand dashboard capabilities to include detailed regional and global environmental impact metrics.
    - Offer personalized user experiences with AI-driven recommendations and real-time notifications.
4. **Material Innovation:**
    - Invest in research and development for biodegradable and eco-friendly materials for fountains and related components.

Future enhancements will ensure adaptability, scalability, and meaningful user engagement, setting a benchmark for eco-friendly innovations. By leveraging AI-driven insights, cutting-edge technologies, and strategic partnerships, the Aqua Cycle Project is positioned for long-term success and global impact. Collaborating with stakeholders, securing funding, and focusing on scaling efforts will further amplify the Aqua Cycle Project’s mission of promoting environmental sustainability.

MEET
THE TEAM

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THE AQUA CYCLE

SCAN AND EXPLORE OUR PROTO TYPE

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APPENDIX

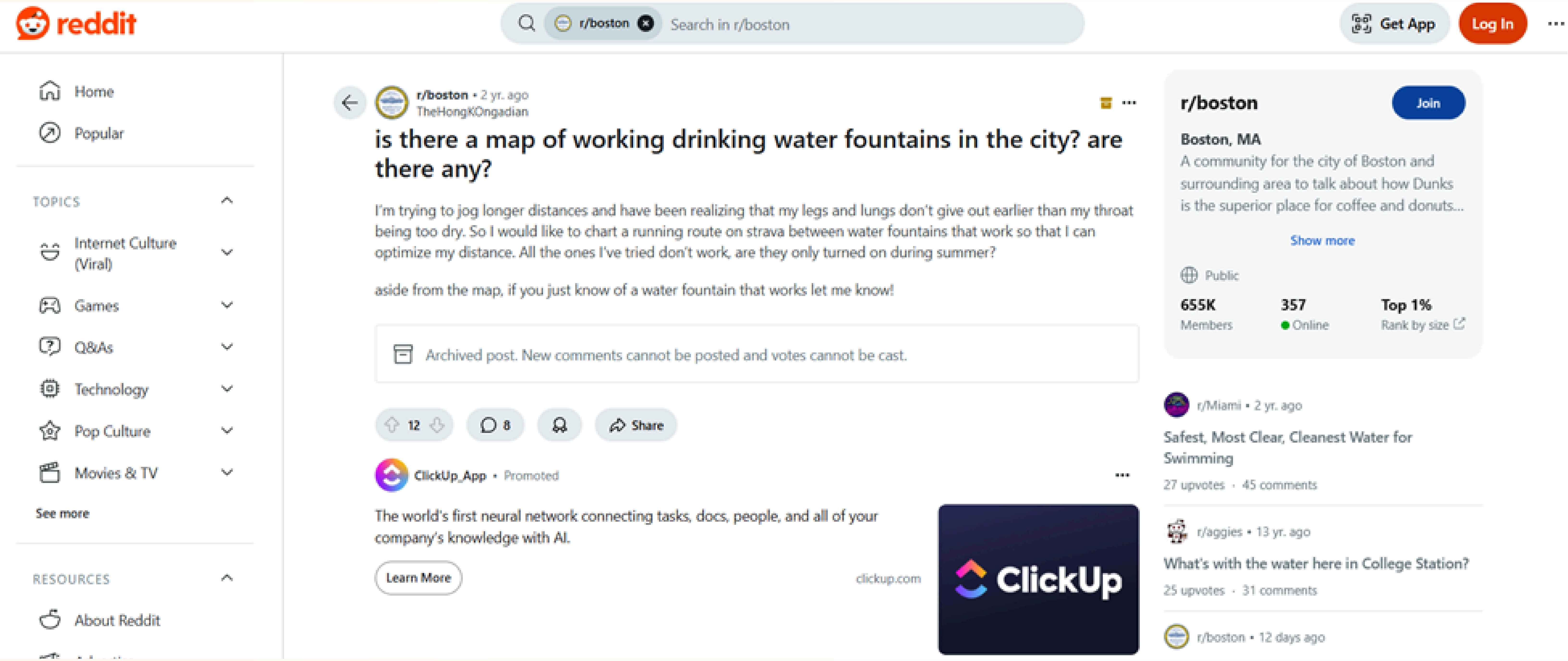
Expected ROI:

Year	Cumulative Revenue	Cumulative Costs	Net Profit	ROI (%)
Year 1	$225,000	$445,000	-$220,000	-52%
Year 2	$495,000	$470,000	$25,000	6%
Year 3	$819,000	$495,000	$324,000	78%
Year 4	$1,208,000	$520,000	$688,000	164%
Year 5	$1,675,000	$545,000	$1,130,000	268%
Year 6	$2,235,000	$570,000	$1,665,000	396%

Cost Structure for 50–60 Smart Water Fountains

Category	Unit Cost (USD)	Cost for 50 Units (USD)	Cost for 60 Units (USD)	Comments
Fountain Installation	$4,000 - $7,000 per unit	$200,000 - $350,000	$240,000 - $420,000	Covers IoT-enabled fountain installation.
Infrastructure Upgrades	$1,000 - $2,000 per unit	$50,000 - $100,000	$60,000 - $120,000	Water and internet upgrades per fountain.
App Development	Fixed	$50,000	$50,000	Basic app with geolocation and dashboard.
AI Predictive Model	Fixed	$20,000	$20,000	AI models for predictive maintenance.
Maintenance (Annual)	$500 per unit	$25,000	$30,000	Cleaning, repairs, and filter replacements.
Community Engagement	Fixed	$20,000	$20,000	Promotional campaigns to raise public awareness.
Reusable Bottle Distribution	$5 per bottle	$25,000 (5,000 bottles)	$30,000 (6,000 bottles)	Cost of distributing reusable bottles.

Category	Unit Cost (USD)	Cost for 10 Units (USD)	Comments
Fountain Installation	$4,000 - $7,000 per unit	$40,000 - $70,000	Covers IoT-enabled fountai installation.
Infrastructure Upgrades	$1,000 - $2,000 per unit	$10,000 - $20,000	Water and internet upgrad fountain.
App Development	$50,000	$50,000	Basic app with geolocation dashboard.
AI Predictive Model	$20,000	$20,000	AI models for predictive ma
Maintenance (Annual)	$500 per unit	$5,000	Cleaning, repairs, and filter replacements.
Community Engagement	-	$10,000	Promotional campaigns to awareness.
Reusable Bottle Distribution	-	$5,000	Distributing reusable bottle encourage use.


reddit
r/boston
Search in r/boston
Get App
Log In
Home
Popular
TOPICS
Internet Culture (Viral)
Games
Q&As
Technology
Pop Culture
Movies & TV
See more
RESOURCES
About Reddit
r/boston • 2 yr. ago
TheHongKOngadian
is there a map of working drinking water fountains in the city? are there any?
I'm trying to jog longer distances and have been realizing that my legs and lungs don't give out earlier than my throat being too dry. So I would like to chart a running route on strava between water fountains that work so that I can optimize my distance. All the ones I've tried don't work, are they only turned on during summer?
aside from the map, if you just know of a water fountain that works let me know!
Archived post. New comments cannot be posted and votes cannot be cast.
12
8
Share
ClickUp_App • Promoted
The world's first neural network connecting tasks, docs, people, and all of your company's knowledge with AI.
Learn More
clickup.com
ClickUp
r/boston
Join
Boston, MA
A community for the city of Boston and surrounding area to talk about how Dunks is the superior place for coffee and donuts...
Show more
Public
655K
Members
357
Online
Top 1%
Rank by size
r/Miami • 2 yr. ago
Safest, Most Clear, Cleanest Water for Swimming
27 upvotes · 45 comments
r/aggies • 13 yr. ago
What's with the water here in College Station?
25 upvotes · 31 comments
r/boston • 12 days ago